

Hareena Chowdary Polavaram

GEN AI Engineer – Generative AI Models, Transformer Architectures & Vector Systems

LinkedIn | 734-510-0652 | hareenachowdary.p@gmail.com | Portfolio | GitHub | USA (Open to Relocation/Remote)

Core Stack: NLP/ML Tools · LLM Fine-Tuning (Dolly/LLaMA2) · Pinecone / Milvus / FAISS · Python · Prompt Design · Statistics

PROFESSIONAL SUMMARY

Innovative and solution-driven GEN AI Engineer with a Master of Science in Computer Science and 5 years of experience architecting, fine-tuning, and operationalizing cutting-edge Generative AI applications and deep learning transformer models. Deep technical expertise in Python, Java, Data Structures, NLP/ML tool technologies (BERT, GPT, Dolly, LLaMA2), and fine-tuning pipelines. Proven track record implementing high-throughput semantic search architectures via leading vector databases (Pinecone, Milvus, FAISS, Pgvector), designing functional prompt engineering solutions, and executing advanced statistical analysis (hypothesis testing, regression, time series, clustering) across large-scale datasets.

TECHNICAL SKILLS

- **NLP/ML & Generative AI Models:** Transformers (GPT, BERT, Dolly, LLaMA2), LLM Fine-Tuning, Prompt Engineering, LangChain, RAG Systems, NLTK
- **Vector DB Implementations:** Pinecone, Milvus, FAISS, Pgvector, Word & Text Embeddings, Semantic Similarity Search, Metadata Filtering
- **Deep Learning & Statistical Analysis:** PyTorch, TensorFlow, Scikit-Learn, Hypothesis Testing, Regression Analysis, Time Series Forecasting, Clustering
- **Programming & Data Structures:** Python (FastAPI, pandas, NumPy), Java, C++, Data Structures & Algorithms, REST APIs, JSON Data Preprocessing
- **Data Engineering & Cloud Platforms:** Microsoft Azure (Certified Data Engineer), GCP, AWS, PostgreSQL, MongoDB, Docker, CI/CD, Git

EXPERIENCE

GEN AI Engineer – NLP/ML Platforms

Jun 2025 – Present

Capital One

Remote, USA

- Develops enterprise GenAI applications utilizing transformer architectures, implementing fine-tuning workflows on Dolly and LLaMA models.
- Builds semantic search and vector retrieval pipelines utilizing Pinecone, Milvus, and FAISS to ground LLM responses with accurate embeddings.
- Designs functional prompt templates, system instructions, and RAG evaluation frameworks in Python to maximize prompt grounding and token efficiency.
- Performs feature engineering and data cleansing on high-dimensional datasets, handling missing values to guarantee downstream data integrity.

AI Automation Engineer – Deep Learning & NLP Systems

May 2024 – May 2025

Advia Credit Union

MI, USA

- Engineered conversational AI systems and NLP text classification pipelines leveraging deep learning models to automate customer workflows.
- Integrated vector database indexing with Pgvector and Pinecone, conducting prompt engineering to enhance real-time query comprehension.
- Executed statistical analyses including hypothesis testing, time series forecasting, and clustering to evaluate model efficacy and drift.
- Automated data ingestion and preprocessing pipelines in Python and SQL across Azure cloud storage, reducing data latency by 40%.

Python Developer – Backend & Data Infrastructure

Aug 2021 – Jul 2023

Tata Consultancy Services

Hybrid, India

- Co-managed enterprise data pipelines and backend services across 30+ countries using Python, Java, SQL, and Azure infrastructure.

- Executed data manipulation and cleansing scripts on large relational tables in PostgreSQL, maintaining robust database integrity and uptime.
- Served as Certified ScrumMaster in Jira, leading sprint planning, hands-on code reviews, and cross-functional technical delivery.

Python Intern – Machine Learning & Data Cleansing

Jun 2020 – Jun 2021

Mphasis

Hybrid, India

- Programmed Python data cleansing scripts to preprocess raw data feeds, perform feature engineering, and handle missing attributes.
- Trained baseline statistical regression and clustering models using Scikit-Learn and NumPy, verifying data pipeline performance.

EDUCATION

Master of Science in Computer Science

Aug 2023 – Apr 2025

Western Michigan University

GPA: 3.92 / 4.0

- Coursework: Artificial Intelligence, Advanced Artificial Intelligence, Machine Learning, Big Data using Python, Distributed Systems, Neural Networks & Deep Learning, Systems Architecture, Statistical Data Modeling

Bachelor of Technology in Computer Science and Engineering

Jul 2017 – Jul 2021

Sri Venkateswara University

GPA: 3.8 / 4.0

CERTIFICATIONS & PUBLICATIONS

- AWS Certified AI Practitioner
- Foundations: Data, Data, Everywhere - Google
- Data Analytics Essentials, Introduction to Data Science - Cisco
- Databases and SQL for Data Science with Python, ETL and Data Pipeline with Shell, Airflow and Kafka - IBM
- “De-Authentication attacks on Rogue UAVs”, ICISS (IEEE), 2021